



Original Article

Evaluation of trypan blue stain in the TC20 automated cell counter as a point-of-care for the enumeration of viable cryptococcal cells in cerebrospinal fluid

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Received 5 March 2017; Revised 17 May 2017; Accepted 18 August 2017; Editorial Decision 26 June 2017

Abstract

Cerebrospinal fluid (CSF) culture can determine a quantitative viability of *Cryptococcus* yeasts; however, culture has a long turnaround-time. The TC20 automated cell counter (Bio-Rad) is a benchtop instrument used to count cells in 30 seconds. *In vitro* studies suggest trypan blue staining can distinguish between viable and dead cryptococcal yeasts. We hypothesized that trypan blue staining with automated cell counting may provide rapid quantification of viable CSF *Cryptococcus* yeasts. In sum, 96 HIV-infected participants with cryptococcal meningitis were enrolled and provided 194 CSF specimens in Kampala, Uganda. Cryptococcosis was diagnosed by CSF cryptococcal antigen (CRAG). CSF was stained with trypan blue and quantified yeasts with the TC20 cell counter. We compared the log₁₀ transformed cell counter readings with gating of 4–10 µm versus log₁₀ quantitative *Cryptococcus* cultures/ml. TC20 showed more positive results (95.4%) overall than culture (78.4%) with reference to CSF CRAG. TC20 had higher readings compared to culture in most cases with only a 25% level of agreement between the two methods. TC20 had a poor correlation to culture throughout the 14 days of antifungal therapy. The median of log₁₀ transformed counts were 5.22 (IQR = 4.79–5.44) for the TC20 and 3.99 (IQR = 2.59–5.14) for culture. Overall, a linear regression showed no significant relationship between the TC20 and culture ($r = -0.0025$; $P = .92$). TC20 automated cell

counting with trypan blue staining was poorly predictive of the quantitative CSF culture and could not be used as a substitute for quantitative culture.

Key words: Cryptococcal infection, HIV, diagnostic techniques, point-of-care systems, quantitation, TC20.

Introduction

Cryptococcal meningitis is a fatal disease accounting for 15–20% of AIDS-related deaths, with the majority occurring in sub-Saharan Africa.^{1–3} Fungal culture is currently the only validated method of determining CSF sterility.⁴ However, culture has a long turnaround-time of up to 10–14 days, which makes culture untenable for real-time clinical decision making for initial meningitis and to differentiate between immune reconstitution inflammatory syndrome (IRIS) and culture-positive relapse.⁵

The TC20 automated cell counter (Bio-Rad Inc, Hercules, California) is a benchtop instrument used to count mammalian cells in a single step. The instrument has an inbuilt autofocus technology and digital image analysis algorithm that provides automated cell counts within 30 seconds. The instrument provides cell counts for suspensions at concentrations within the range of 5×10^4 to 1×10^7 cells/ml and for cells within the range of 6–50 μm cell diameter. When trypan blue is included, the TC20 can automatically detect and assesses cell viability via trypan blue exclusion. For samples with mixed cell populations, the user can define the cell population of interest by adjusting the cell size of interest by gating. Previous studies have shown that the TC20 performance is comparable or even better than the hemocytometer with manual counting.⁶ Similarly, recent studies suggest utility of trypan blue stain in distinguishing between viable and dead *Cryptococcus* yeasts.⁷ In the current study, we used the TC20 to count viable *Cryptococcus* yeasts in fresh whole CSF in comparison to quantitative fungal cultures in a clinical setting. We hypothesized that the TC20 may give automated counts, which may be approximately predictive of the quantitative CSF culture result when used with trypan blue staining.

Methods

Study population and eEthical statement

The study population included sequential persons diagnosed with cryptococcal meningitis by CSF cryptococcal antigen (CRAG) lateral flow assay (Immy, Norman, Oklahoma) at Mulago National Referral Hospital in Kampala, Uganda, during the “Adjunctive Sertraline for the Treatment of HIV-Associated Cryptococcal Meningitis” (ASTRO-CM) pilot study.⁸ Participants included in the

study were ≥ 18 years, infected with human immunodeficiency virus (HIV), with a positive CSF CRAG. CSF was collected at days 0, 3, 7, 10, 14, and as clinically indicated by therapeutic lumbar punctures as part of standard care for the control of raised intracranial pressure. Participants or their surrogate provided written informed consent. Ethical approval was obtained from the Uganda National Council of Science and Technology (UNCST), Makerere University Institutional Review Board, Mulago Hospital Research and Ethics Committee, and University of Minnesota.

Study procedures

In this current study, ASTRO participants with a positive CSF CRAG were included after written informed consent. Symptomatic HIV-infected persons being managed for cryptococcal meningitis on the infectious disease ward had lumbar punctures performed at serial time points. CSF was stained with trypan blue and quantified yeasts with the TC20 automated cell counter to estimate the number of viable cryptococcal cells per milliliter subtracting the number of CSF white cells per milliliter. Fungal cultures on whole CSF were performed and incubated at 30°C for up to 10 days on Sabouraud dextrose agar.⁹ Cultures were recorded as colony forming units per milliliter (cfu/ml). The number of viable *Cryptococcus* yeasts per milliliter in the TC20 was compared to quantitative number of cfu/ml by fungal culture.

Preliminary validation tests/studies

Before comparing the TC20 to the gold standard (culture), cell counts were performed on randomly selected CRAG positive CSF samples to estimate the optimal size/gating for most yeast cells. Repeatability tests were performed for precision, recording four observations for each sample by the same reader and machine without removing the slide from the machine. These precision experiments were performed “without gating” ($n = 10$) and with a gating of 4–10 μm ($n = 50$). In addition, fungal medium was prepared containing Sabouraud dextrose agar, chloramphenicol, and trypan blue stain. Known cryptococcal isolates were cultured on this medium to check whether viable yeast cells would take up the trypan blue stain.

TC20 automated counting

For this procedure, a 1:2 dilution of fresh, whole CSF was made in 0.4% trypan blue stain in a cryovial (10 µl of CSF added to 10 µl of trypan blue). Ten microliters of the mixture were then loaded into the opening of the TC20 counting slide. The slide was then inserted into the TC20 instrument. The cell counter automatically detects the presence of the counting slide and initiates the count. A gating of 4–10 µm was selected as optimal for cryptococcal cells based on preliminary validation studies performed. Cells were counted within 5 minutes of mixing trypan blue. Images of cells counted were automatically saved on a USB drive inserted in the machine. Captured images were later analyzed using the TC20 data analyzer software.

Quantitative fungal cultures

Quantitative cryptococcal cultures were performed on whole CSF using four 10-fold serial dilutions and the growth recorded as CFU/ml.⁹ Cryptococcal culture plates were incubated at 30°C for up to 10 days on Sabouraud dextrose agar with chloramphenicol.

Statistical analysis

Statistical analysis was aimed at establishing the reliability, level of agreement and deviation of TC20 (viable cells/ml) compared to quantitative cryptococcal cultures (CFU/ml) as well as the determination of CSF sterility at 95% confidence interval (CI). Data were analyzed using STATA version 13 (STATA, College Station, Texas).

Results

Study population characteristics

Between March 2015 and December 2015, we enrolled 96 participants with cryptococcal meningitis who provided 194 CSF specimens from both diagnostic and therapeutic lumbar punctures. Cryptococcal meningitis was diagnosed by CSF CRAG lateral flow assay. Of participants, 39% (37/96) were women, and the median age of all participants was 34.5 years (interquartile range [IQR], 28–40). All participants were HIV-infected adults with a median CD4 T cell count of 14 cells/µl (IQR, 5–51; max 235 cells/µl; $n = 92$) at diagnosis. Only 51% (49/96) of the participants were receiving antiretroviral therapy, and 5.2% (5/96) had defaulted from antiretroviral therapy at diagnosis. Among patients reporting a headache ($n = 92$), the median headache duration was 14 days (IQR, 7–30; max 60 days). The CSF opening pressures at baseline had a median of 322 mmH₂O (IQR = 210–440; max = 930 mmH₂O). Among partici-

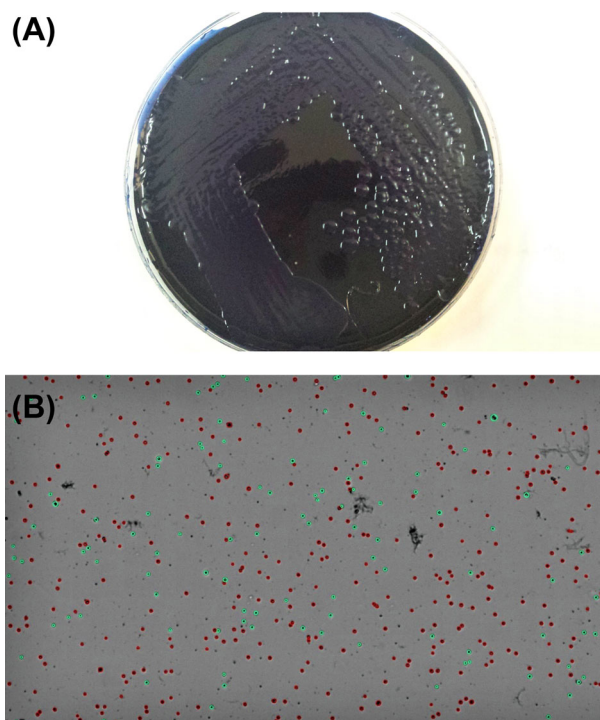


Figure 1. (A) This is a culture plate with cryptococcal isolates grown on medium containing Sabouraud dextrose agar, chloramphenicol and trypan blue. The colonies appear dark blue. (B) This is an image taken by the TC20 and analysed with the data analyser. The dead yeast cells appear red while the live cells appear green. This Figure is reproduced in color in the online version of *Medical Mycology*.

pants, 67% (60/90) had < 5 white cells/µl in CSF at diagnosis.

Distinguishing *Cryptococcus* viability

At the beginning of the study, we incubated three cryptococcal isolates (0.5 McFarland concentration) at 55°C for 24 hours to heat kill the yeast cells. The heat killed cells were then treated with 0.4% trypan blue stain. The heat killed cells took up the stain, while yeasts in similar preparations left at room temperature did not take up the trypan blue stain. Similarly, we treated one cryptococcal isolate with amphotericin B for 24 hours to kill the yeasts with an antifungal agent. The isolate treated with amphotericin B, when mixed with trypan and observed under a light microscope, showed that 98% of the cells were live and not killed by the antifungal treatment. Medium was prepared containing Sabouraud dextrose agar, chloramphenicol, and trypan blue on which we subcultured cryptococcal isolates. These grew dark blue colonies (Fig. 1A). However, when viewed in a wet preparation, it showed that all the cells from the dark blue colonies were viable, and none had taken up the stain.

Determining the optimal gating for yeast counting

Without the user defining the cell population of interest, the instrument automatically decides what population of cells to count. Therefore, we first performed cell counts on randomly selected CRAG positive CSF samples ($n = 16$) without defining the population of interest. On the same samples, cells were counted using a gating of 4–10 μm and 4–25 μm . We observed no difference in the readings for the 4–10 μm and 4–25 μm gatings. Using the TC20 data analyzer software, it was discovered that for all gatings, there were cell populations ranging in 5–20 μm . However, for each sample, more than 95% of the yeast cells had a diameter of 5–7 μm . Based on the same software, each sample had more dead cells than live cells. In some images, cells beyond 10 μm counted seemed to be debris (Fig. 1B). Therefore, in all preceding validation tests, we decided to use a gating of 4–10 μm as the optimal size for cryptococcal cells.

Repeatability and precision

Repeatability tests were performed, recording four observations (live cell count) for each sample by the same reader without removing the slide from the instrument “without gating” ($n = 10$). A repeated measures ANOVA analysis indicated a significant correlation among the four readings within each sample ($r = 0.9721$, $P < .001$). We then tested for random effects, and there was no significant difference among the four readings within each sample ($P = .63$). Similarly, repeatability tests were performed, recording four observations (live cell count) for each sample by the same reader without removing the slide from the instrument, using a gating of “4–10 μm ” ($n = 50$). A repeated measures ANOVA analysis indicated a significant correlation among the four readings within each sample ($r = 0.9499$; $P < .001$). We then tested for random effects, and there was a borderline significant difference among the four readings within each sample ($P = .045$).

TC20 versus quantitative fungal cultures

$\log_{10}$ transformed readings of the TC20 (live cells/ml) were compared at a gating of 4–10 μm to quantitative fungal cultures (gold standard) (CFU/ml). Overall, the median of $\log_{10}$ transformed counts were 5.22 ($n = 178$; IQR = 4.79–5.44; max = 6.15) for the TC20 and 3.99 ($n = 151$; IQR = 2.59–5.14; max = 7.71) for positive cultures. Among the 194 CRAG positive CSF samples which were longitudinally collected during antifungal therapy, TC20

had more positive readings (95.4%) than culture (78.4%). There was no statistical correlation between the TC20 reading and quantitative culture at day 1 ($R^2 = 0.39$, $P = .77$), day 3 ($R^2 = 0.066$, $P = .79$), day 7 ($R^2 = -0.025$, $P = .96$), or day 14 ($R^2 = -0.27$, $P = .61$) of amphotericin B combination therapy (Fig. 2). There was a significant difference in the means of counts between TC20 and culture (difference 1.30, 95% CI 1.03 to 1.58 $\log_{10}$ CFU/ml, $P < .001$). Even among day 1 readings, among persons with CSF < 5 white cells/ μl ($n = 23$), the correlation was poor ($R^2 = 0.3703$, $P = .082$). Furthermore, there was no significant association between the TC20 readings and quantitative culture by linear regression ($r = -0.0025$; 95% CI, -0.0513 to 0.4633 ; $P = .92$). We then compared the yeast counts at the initial diagnostic lumbar puncture versus the follow up counts at time of therapeutic lumbar punctures. With amphotericin therapy, quantitative CSF cultures declined over time ($P < .001$). Yet the mean automated TC20 cell counts did not decline between the initial diagnostic CSF specimen and other therapeutic lumbar punctures even with use of the trypan blue staining ($P = .88$). When compared to the baseline CSF opening pressures ($n = 54$), there was no significant relationship between the TC20 and opening pressure ($P = .88$). However, there was a significant relationship between culture and opening pressure ($R^2 = 0.34$, $P = .011$, 95% CI).

Discussion

The TC20 automated cell counter did not predict the outcome of quantitative cell culture. In the current study, we worked on the principle that viable cryptococcal yeasts do not take up trypan blue stain, while dead yeasts take up the stain based on our initial validation tests. This is supported by a recent publication that used *C. neoformans* var. *grubii* strain H99.⁷ However, we could not explain why cells treated with amphotericin B remained viable at 24 hours. Another paper published in 1974 used fungal medium with antibiotics and trypan blue to culture *Cryptococcus* and *Candida albicans*. *Cryptococcus* showed dark blue colonies (indicating uptake of the stain by the cells), while *C. albicans* showed light/white colonies. The authors concluded that viable cryptococcal yeasts can take up the trypan blue stain.¹⁰ This would explain the high percentage of dead cells recorded in each sample using the TC20. However, when we reproduced this experiment in our laboratory, microscopy results of the dark blue colonies in a wet preparation showed that all the cells were viable and none had taken up the stain. It seemed that the stain just made a top coating, but the viable cryptococcal cells did not take up the trypan blue stain.

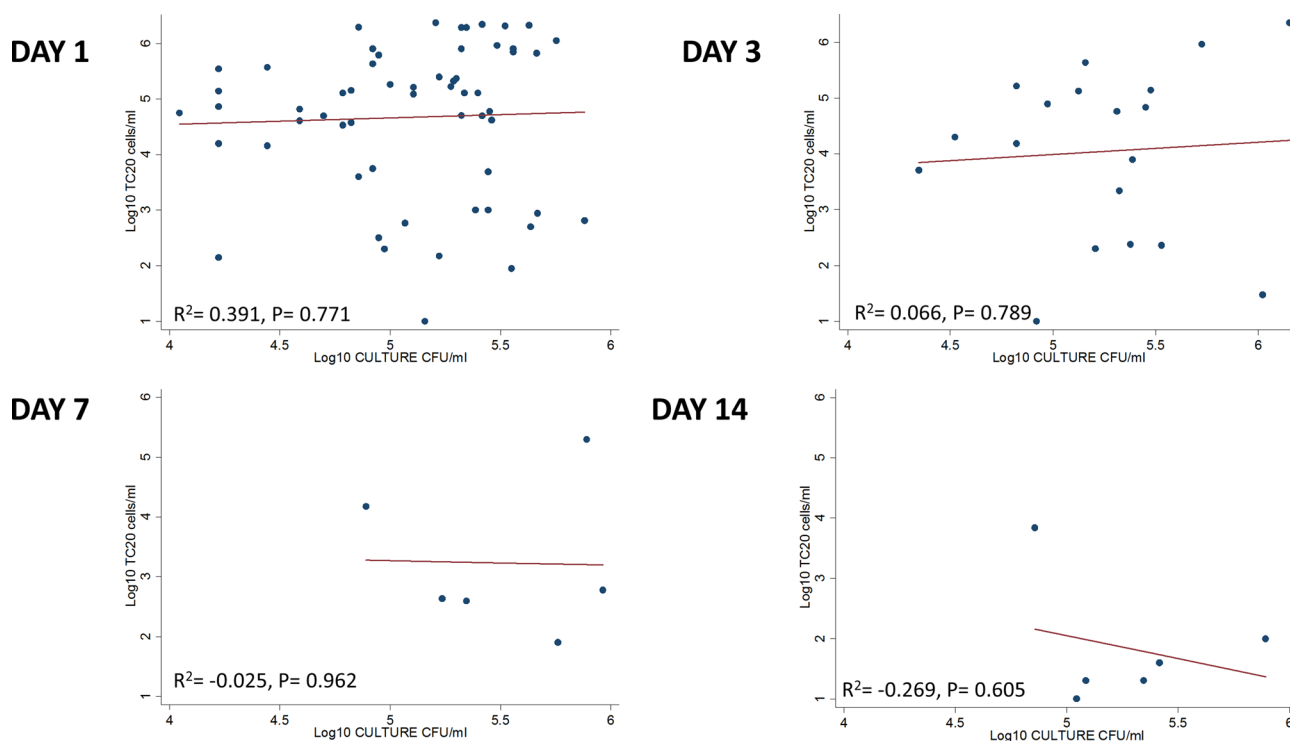


Figure 2. Scatter plots of TC20 automated cell count vs CSF quantitative culture from days 1, 3, 7, and 14 during antifungal therapy. The correlation worsened over time from day 1 to 14 ($R^2 = 0.39, 0.066, -0.025, -0.27$), respectively, and was non-statistically significant. This Figure is reproduced in color in the online version of *Medical Mycology*.

Based on our analysis with the TC20 data analysis software, there were cell populations ranging in 5–20 μm . This agrees with most literature that gives a wide range of the cell sizes typical of *Cryptococcus*.¹¹ This is why in our preliminary tests we used a gating of 4–10 μm , 4–25 μm , and without gating. The TC20 instrument had a lower limit of 4 μm , so the 4–10 μm gating was chosen as optimal for most yeast cells using the TC20 since more than 95% of the yeast had a cell body diameter of 5–7 μm (excluding the capsule). Previous work in a similar population at Mulago Hospital showed an *ex vivo* total cell diameter of 16.1 μm (IQR = 10.3–29.8; $n = 122$) for cryptococcal cells including the capsule.¹¹ Analysis of images from the TC20 machine showed that each sample had more dead cells than live cells. In some images, counted cells that were bigger than 10 μm seemed to be debris but not yeast cells. We assumed that some white blood cells also fell in the same category. Similarly, some cells were not counted by the instrument. They were probably read as debris. To eliminate the effect due to white blood cells, we tried to increase the gating to 20–50 μm . We also tried a 1:10 dilution of the CSF samples. However, these two procedures did not give any useful information and all counts were zero.

Repeatability tests on the TC20 showed that there were no significant differences among the four readings, with a high consistence. When compared to quantitative cryp-

tococcal cultures, it showed that the TC20 had higher readings compared to culture in most cases with only 25% of the samples having the exact count between the two methods. However, there was a statistically significant difference ($P = .001$) between the two methods. We found a weak correlation between the TC20 and culture which worsened with increasing doses of amphotericin therapy. Only culture had a significant relationship with CSF opening pressures at baseline ($P = .011$). Similarly, culture ($P < .001$) had a significant difference between the means of counts at baseline and other therapeutic lumbar punctures. Among the samples with no culture growth ($n = 42$), only six (14.3%) had zero counts with the TC20. The TC20 counts ranged in 0–316,000 (cells/ml) among these 42 samples. A recently published bulletin (Bio-Rad bulletin 6003) compared the TC20 to the hemocytometer using HeLa cells; and reported that automated cell counting could significantly reduce both user and concentration-dependent variances while reducing the turn-around time.⁶

Study limitations

The major limitation to the study that we could not overlook was that the instrument may not have the ability to distinguish between yeast cells and other related cells or debris of the same size. However, since we were dealing

with a sterile fluid collected aseptically, we assume the effect due to this was minimal. In addition, cells that had a diameter less than 4 μm could have been missed since the lower limit of the machine was 4 μm . This cell size would be uncommon for *Cryptococcus*.¹¹ Based on the analyzed images, some cells were actually not counted by the instrument, probably seen as debris and/or confused by the capsule. This could also have affected the results. Only trypan blue viability was tested with the TC20 cell counter and it remains possible that other viability stains may work with the TC20.

In conclusion, the TC20 automated cell counter was not a replacement for quantitative culture. More studies are needed to explore the utility of simple, low cost methods to rapidly determine fungal burden which can guide therapy more rapidly than waiting for quantitative culture growth.

ASTRO-CM Team members Rueben Kiggundu, Liliane Tugume, Mahsa Abassi, Edward Mpoza, Henry W Nabeta, Jane Francis Ndyetukira, Cynthia Ahimbisibwe, Florence Kugonza, Alisat Sadiq, Tony Luggya, Julian Kaboggoza, Katelyn Pastick, Eva Laker, Elissa K Butler, Jonathan Dual, A. Wendy Fujita, Nathan Yueh, Alice Namudde, Ryan Halupnick, Bilal Jawed, Liliane Mukaremera, and Kirsten Nielsen.

Acknowledgments

We thank institutional support from Drs. Richard Brough, Rosalind Parkes Ratanshi and Mohammed Lamorde. This work was supported by the National Institutes of Health (R01NS086312, T32AI055433 R25TW009345, K01TW010268), Grand Challenges Canada (S4 0296-01), and United Kingdom Medical Research Council (MR/M007413/1).

Authors' contributions

Conceived and designed concept: RK, AA, DRB. Performed experiments: RK, AA, TKK. Analysed data: RK, MSN, DRB. Contributed reagents/materials/analysis tools: RK, DBM, AK, JR, DRB. Participated in initial manuscript drafting: RK, MSN. Participated in critical revisions for intellectual content: RK, DW, AK, DBM, JR, DRB. Participated in obtaining funding: DBM, DRB. Participated in administrative, technical, or material support: DW, DBM, JR.

Declaration of interest

The authors report no conflicts of interest. The authors alone are responsible for the content and the writing of the paper.

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